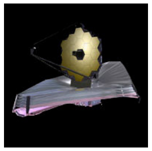

8762 - Late-Time Near/Mid-Infrared Nebular Spectroscopy of a Kilonova with JWST

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	Epoch 1 - NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) KN
	2	Epoch 1 -MIRI Spectroscopy	MIRI Low Resolution Spectroscopy	(1) KN
	3	Epoch 2 - NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) KN
	4	Epoch 2 - MIRI Spectroscopy	MIRI Low Resolution Spectroscopy	(1) KN
	5	Epoch 3 - NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) KN

Folder	Observation	Label	Observing Template	Science Target
	6	Epoch 3 - MIRI Photometry	MIRI Imaging	(1) KN

ABSTRACT

This proposal seeks to obtain the first observational evidence that compact object mergers involving a neutron star are capable of producing the third-peak r-process elements as well as characterize the diversity of kilonova elemental abundances to constrain kilonova modeling. The kilonova will be detected and identified by an optical or infrared sky survey, and spectroscopically confirmed. The observations consist of three epochs from approxiamtely 25 to 70 days post-merger: (1) NIRSPEC + MIRI spectroscopy, (2) NIRSPEC + MIRI spectroscopy, and (3) NIRSPEC spectroscopy + MIRI imaging.

These observations will: (i) obtain spectroscopy with high SNR for a broad wavelength range to detect and identify as many spectral features as possible, (ii) precisely measure the full infrared spectral energy distribution to inform theoretical ejecta models, identify ejecta mechanisms, and produce the first MIR spectroscopy of a kilonova, and (iii) measure the change in bolometric luminosity between epochs to reveal dominant radioactive isotopes in the ejecta and constrain the presence of the heaviest elements on the periodic table.

We waive our proprietary rights to enable the whole community to benefit from these data.

OBSERVING DESCRIPTION

The observations consist of three epochs of a spectroscopically confirmed kilonova at late time, likely discovered by an optical or infrared sky survey. There are a total of six observations across three epochs:

Epoch 1: ~25-35 days post merger

This requires a non-disruptive ToO.

Two components grouped together:

  NIRSPEC low-resolution spectroscopy with the S200A1 fixed slit

  MIRI low-resolution slit spectroscopy

Epoch 2: ~40-50 days post merger

5-25 days after the first epoch.

Two components grouped together:

NIRSPEC low-resolution spectroscopy with the S200A1 fixed slit

MIRI low-resolution slit spectroscopy

Epoch 3: ~60-70 days post merger

25-45 days after the first epoch.

Two components grouped together:

NIRSPEC low-resolution spectroscopy with the S200A1 fixed slit

MIRI imaging in the F560W and F770W filters

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	
(2)	Reference-Star	Bright IR target acquisition star	

Comments: Assuming reference star is 18th mag in K band

Observation

Proposal 8762, Observation 1: Epoch 1 - NIRSPEC

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 Reference-Star	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	222188

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	2	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	100	8	1	NONE	4	32	5036.111	222188

Special Requirements	
	Target Of Opportunity Response Time 21 Days, Carry-Over 3 After 1 by 5 Days to 25 Days 4 After 1 by 5 Days to 25 Days 5 After 1 by 25 Days to 45 Days 6 After 1 by 25 Days to 45 Days Group Observations 1, 2, Non-interruptible

Observation

Proposal 8762, Observation 2: Epoch 1 -MIRI Spectroscopy

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 Reference-Star	F560W	FAST	4	1	1	11.1	222188

Template

Subarray	Obtain Verification Image?
FULL	false

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	100	12	24	1	2	6721.147	222188

Special Requirements	
	Target Of Opportunity Response Time 21 Days, Carry-Over Group Observations 1, 2, Non-interruptible

Observation

Proposal 8762, Observation 3: Epoch 2 - NIRSPEC

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 Reference-Star	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	222188

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	2	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	100	8	1	NONE	4	32	5036.111	222188

Special Requirements	
	On Hold On Hold for after 5-25 days after the first epoch 3 After 1 by 5 Days to 25 Days Group Observations 3, 4, Non-interruptible

Observation

Proposal 8762, Observation 4: Epoch 2 - MIRI Spectroscopy

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 Reference-Star	F560W	FAST	4	1	1	11.1	222188

Template

Subarray	Obtain Verification Image?
FULL	false

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	125	20	40	1	2	13980.652	222188

Special Requirements	
	On Hold On hold for after 5-25 days after epoch 1 4 After 1 by 5 Days to 25 Days Group Observations 3, 4, Non-interruptible

Observation

Proposal 8762, Observation 5: Epoch 3 - NIRSPEC

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 Reference-Star	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	222188

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	2	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	100	8	1	NONE	4	32	5036.111	222188

Special Requirements	
	On Hold On Hold for 25-45 days after the first epoch 5 After 1 by 25 Days to 45 Days Group Observations 5, 6, Non-interruptible

Observation

Proposal 8762, Observation 6: Epoch 3 - MIRI Photometry

Fri Oct 17 21:00:13 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	KN	Spectroscopically confirmed KN brighter than	

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	2-Point								DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	50	4	1	Dither 1	2	8	1126.666	222188
2	F770W	FASTR1	75	12	1	Dither 1	2	24	5056.123	222188

Special Requirements

On Hold On Hold for 25-45 days after the first epoch

6 After 1 by 25 Days to 45 Days
Group Observations 5, 6, Non-interruptible